

Customer Shopping Behaviour Analysis

Project Report

This report presents a complete end-to-end Data Analyst project focused on Customer Shopping Behaviour Analysis. The project includes data cleaning using Python, exploratory data analysis (EDA), SQL-based business problem solving, and dashboard creation in Power BI. The objective of the project is to identify customer trends, purchasing patterns, sales performance, and customer preferences to support business decision-making.

1. About the Dataset

The dataset contains customer shopping behaviour information such as customer demographics, purchase details, product categories, review ratings, payment methods, and purchase frequency. The dataset was used to analyze customer purchasing behaviour and business performance.

2. Data Dictionary

Column Name	Description
customer_id	Unique ID assigned to each customer
age	Age of customer
gender	Gender of customer
category	Product category purchased
item_purchased	Specific item purchased
purchase_amount	Amount spent by customer
review_rating	Customer review rating
payment_method	Method used for payment
subscription_status	Subscription membership status
frequency_of_purchases	Frequency of customer purchases
location	Customer location/city
season	Season in which purchase occurred
color	Preferred product color
size	Product size selected
previous_purchases	Number of previous purchases

3. Data Cleaning Process Using Python

The dataset contained missing values, inconsistent records, duplicate values, and formatting issues. Python and Pandas libraries were used to clean and prepare the dataset before analysis.

- **Correct Mapping of Item Purchased and Category Columns**

Fix category and item purchased column

```
correct_mapping = {  
    #Clothing  
    "Skirt" : "Clothing", "Socks" : "Clothing", "Blouse" : "Clothing", "Shirt" : "Clothing", "Hoodie": "Clothing", "Dress" : "Clothing", "Pant"  
    #Accessories  
    "Scarf" : "Accessories", "Belt" : "Accessories", "Handbag" : "Accessories", "Hat" : "Accessories", "Sunglasses" : "Accessories", "  
    #Footwear  
    "Boots" : "Footwear", "Sandals" : "Footwear", "Shoes" : "Footwear", "Sneakers" : "Footwear",  
    #Outerwear  
    "Coat" : "Outerwear", "Jacket" : "Outerwear"  
}
```

correct_mapping

```
{'Skirt': 'Clothing',  
'Socks': 'Clothing',  
'Blouse': 'Clothing',  
'Shirt': 'Clothing',  
'Hoodie': 'Clothing',  
'Dress': 'Clothing',  
'Pants': 'Clothing',  
'Jeans': 'Clothing',  
'Shorts': 'Clothing',  
'Sweater': 'Clothing',  
'T-shirt': 'Clothing',  
'Scarf': 'Accessories',  
'Belt': 'Accessories',  
'Handbag': 'Accessories',  
'Hat': 'Accessories',  
'Sunglasses': 'Accessories',  
'Jewelry': 'Accessories',  
'Gloves': 'Accessories',  
'T-Shirt': 'Accessories',  
'Boots': 'Footwear',  
'Sandals': 'Footwear',  
'Shoes': 'Footwear',  
'Sneakers': 'Footwear',  
'Coat': 'Outerwear',  
'Jacket': 'Outerwear'}
```

```
df['Category'] = df['item_purchased'].map(correct_mapping)
```

```
df.groupby('Category')['item_purchased'].unique()
```

```
Category  
Accessories          [Scarf, Belt, Handbag, Hat, Sunglasses, Jewelry, Gloves]  
Clothing             [Skirt, Socks, Blouse, Shirt, Hoodie, Dress, Pants, Jeans, Shorts, Sweater, T-shirt]  
Footwear             [Boots, Sandals, Shoes, Sneakers]  
Outerwear            [Coat, Jacket]  
Name: item_purchased, dtype: object
```

- **Treating Missing Values**

Treating the null values

```
df.isnull().sum()
```

```
customer_id      0  
age              0  
gender           0  
item_purchased   0  
category         0  
purchase_amount_(usd)  561  
location         0  
size            377  
color           0  
season          0  
review_rating    608  
subscription_status  0  
discount_applied  0  
previous_purchases  551  
payment_method   0  
frequency_of_purchases  0  
Category         146  
dtype: int64
```

(a) Size Column

Handling missing values of size column using category-based imputation

```
df['size'].unique()
<StringArray>
['M', nan, 'S', 'L', 'XL']
Length: 5, dtype: str
```

```
df.groupby('category')['size'].unique()
category
Accessories      [S, L, M, XL, nan]
Clothing         [M, nan, S, L, XL]
Footwear         [M, nan, XL, L, S]
Outerwear        [M, nan, S, XL, L]
Name: size, dtype: object
```

```
df.loc[df['category'] == 'Accessories', 'size'] = "Free Size"
```

```
df.groupby('category')['size'].unique()
category
Accessories      [Free Size]
Clothing         [M, nan, S, L, XL]
Footwear         [M, nan, XL, L, S]
Outerwear        [M, nan, S, XL, L]
Name: size, dtype: object
```

```
df.loc[df['category'] == 'Footwear', 'size'] = "Not Available"
```

```
df.loc[(df['category'] == 'Clothing') & (df["size"].isnull()), "size"] = df[df["category"] == "Clothing"]["size"].mode()[0]
```

```
df.loc[(df['category'] == 'Outerwear') & (df["size"].isnull()), "size"] = df[df["category"] == "Outerwear"]["size"].mode()[0]
```

```
df.groupby('category')['size'].unique()
category
Accessories      [Free Size]
Clothing         [M, S, L, XL]
Footwear         [Not Available]
Outerwear        [M, S, XL, L]
Name: size, dtype: object

df['size'].isnull().sum()
np.int64(0)
```

(b) Category Column

Treating Category Column

```
df['category'] = df['category'].fillna(df['category'].mode()[0])
```

(c) Review Rating Column

Treating Review Rating Null Values

```
df['review_rating'].mean()

df['review_rating'].median()

df['review_rating'].mode()

df.groupby("item_purchased")['review_rating'].mean()

df['review_rating'] = df['review_rating'].fillna(df.groupby('item_purchased')['review_rating'].transform('mean'))

df.isnull().sum()

df['review_rating'] = df['review_rating'].round(2)
```

(d) Previous Purchases Column

```
Treating Previous Purchases

df['previous_purchases'] = df['previous_purchases'].fillna(0)

df.isnull().sum()
```

(e) Purchase Amount(USD) Column

```
Treating Purchase Amount(USD) Column

df.groupby("item_purchased")["purchase_amount_(usd)"].mean()

Filling Null Values of Purchase Amount (USD) with the mean of each Item Purchased

urchase_amount_(usd)'] = df['purchase_amount_(usd)'].fillna(df.groupby('item_purchased')['purchase_amount_(usd)'].transform('mean'))

df.columns[df.columns.duplicated()]

df = df.loc[:,~df.columns.duplicated()]

df.isnull().sum()
```

- Checking and Removing Duplicates

```
Checking Duplicates

df.duplicated().sum()

Deleting Duplicate Records

df = df.drop_duplicates(subset=['customer_id'], keep = 'first')

df.duplicated().sum()

df.columns
```

- Renaming purchase_amount_(USD) column

```
Renaming Purchase Amount(USD) to purchase_amount

df = df.rename(columns = {"purchase_amount_(usd)": "purchase_amount"})

df.columns
```

- Boolean Mapping of Subscription Status and Discount Applied Columns

Boolean Mapping of Subscription Status and Discount Applied Columns

```
df["subscription_status"] = df["subscription_status"].map({"Yes": True, "No": False})
df["discount_applied"] = df["discount_applied"].map({"Yes": True, "No": False})
```

4. SQL Business Questions

SQL was used to answer important business questions and generate insights from the cleaned dataset.

1. Which product category generates the highest Sales?

	category character varying (50) 🔒	total_sales numeric 🔒
1	Clothing	113069.35
2	Accessories	64463.43
3	Footwear	36006.79
4	Outerwear	19355.96

2. Which gender contributes more to overall sales?

	gender character varying (10) 🔒	total_sales numeric 🔒
1	Male	157875.26
2	Female	75020.27

3. How are customers distributed across different age groups?

	age_group text 🔒	customer_count bigint 🔒
1	50+	1476
2	41-50	750
3	21-30	737
4	31-40	725
5	0-20	212

4. Which season records the highest sales?

	season character varying (20) 🔒	total_sales numeric 🔒
1	Fall	59521.56
2	Spring	58707.99
3	Winter	58670.55
4	Summer	55995.43

5. Which cities have the highest number of customers?

	location character varying (100) 🔒	customer_count bigint 🔒
1	Montana	96
2	California	95
3	Idaho	93
4	Illinois	92
5	Alabama	89

6. Which payment method is most preferred by customers?

	payment_method character varying (50) 🔒	total_customers bigint 🔒
1	PayPal	677
2	Credit Card	671
3	Cash	670
4	Debit Card	636
5	Venmo	634
6	Bank Transfer	612

7. What percentage of customers are subscribed?

	subscription_status boolean 🔒	total_customers bigint 🔒
1	false	2847
2	true	1053

8. Which product categories receive the highest average customer review ratings?

	category character varying (50) 🔒	avg_review_rating numeric 🔒
1	Outerwear	3.86
2	Clothing	3.77
3	Accessories	3.72
4	Footwear	3.72

9. Which purchase frequency is most common among customers?

	frequency_of_purchases character varying (50) 🔒	customer_count bigint 🔒
1	Every 3 Months	584
2	Annually	572
3	Quarterly	563
4	Monthly	553
5	Bi-Weekly	547
6	Fortnightly	542
7	Weekly	539

10. Which product size is most preferred by customers?

	size character varying (20) 🔒	total_customers bigint 🔒
1	M	1107
2	Free Size	1097
3	Not Available	599
4	L	536
5	S	343
6	XL	218

11. Which product colors are most preferred?

	color character varying (30) 🔒	total_customers bigint 🔒
1	Olive	177
2	Yellow	174
3	Silver	173
4	Teal	172
5	Green	169

12. Which payment methods generate the highest sales revenue ?

	payment_method character varying (50) 🔒	total_sales numeric 🔒
1	PayPal	40223.42
2	Credit Card	40222.87
3	Cash	39826.95
4	Debit Card	38607.23
5	Venmo	37435.32

13. Which age groups contribute the highest sales revenue?

	age_group text 🔒	total_sales numeric 🔒
1	50+	88701.28
2	21-30	44540.62
3	41-50	43830.78
4	31-40	43298.56
5	0-20	12524.29

14. How many total customers are present in the dataset?

	total_customers bigint 🔒
1	3900

15. How many unique products/items are available?

	unique_items bigint 🔒
1	25

16. What is the average customer review rating?

	avg_review_rating numeric 🔒
1	3.76

17. What is the total sales revenue generated by the business?

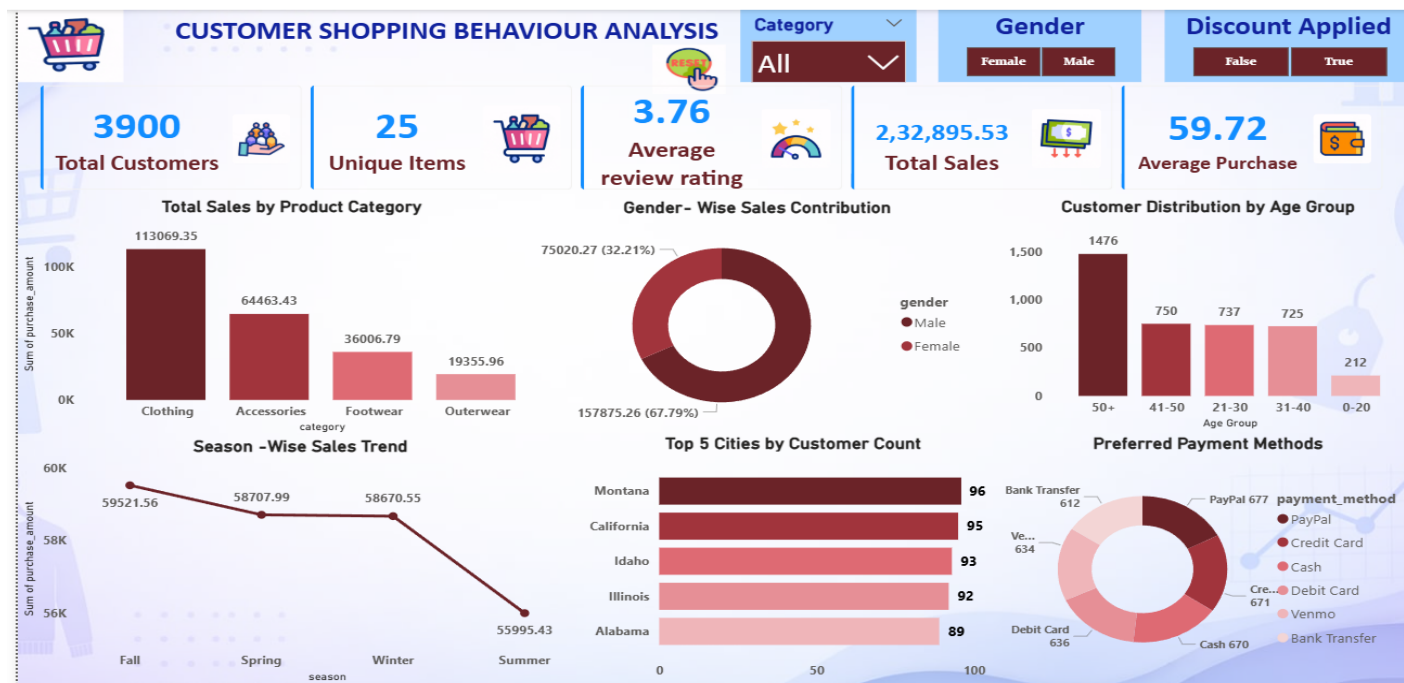
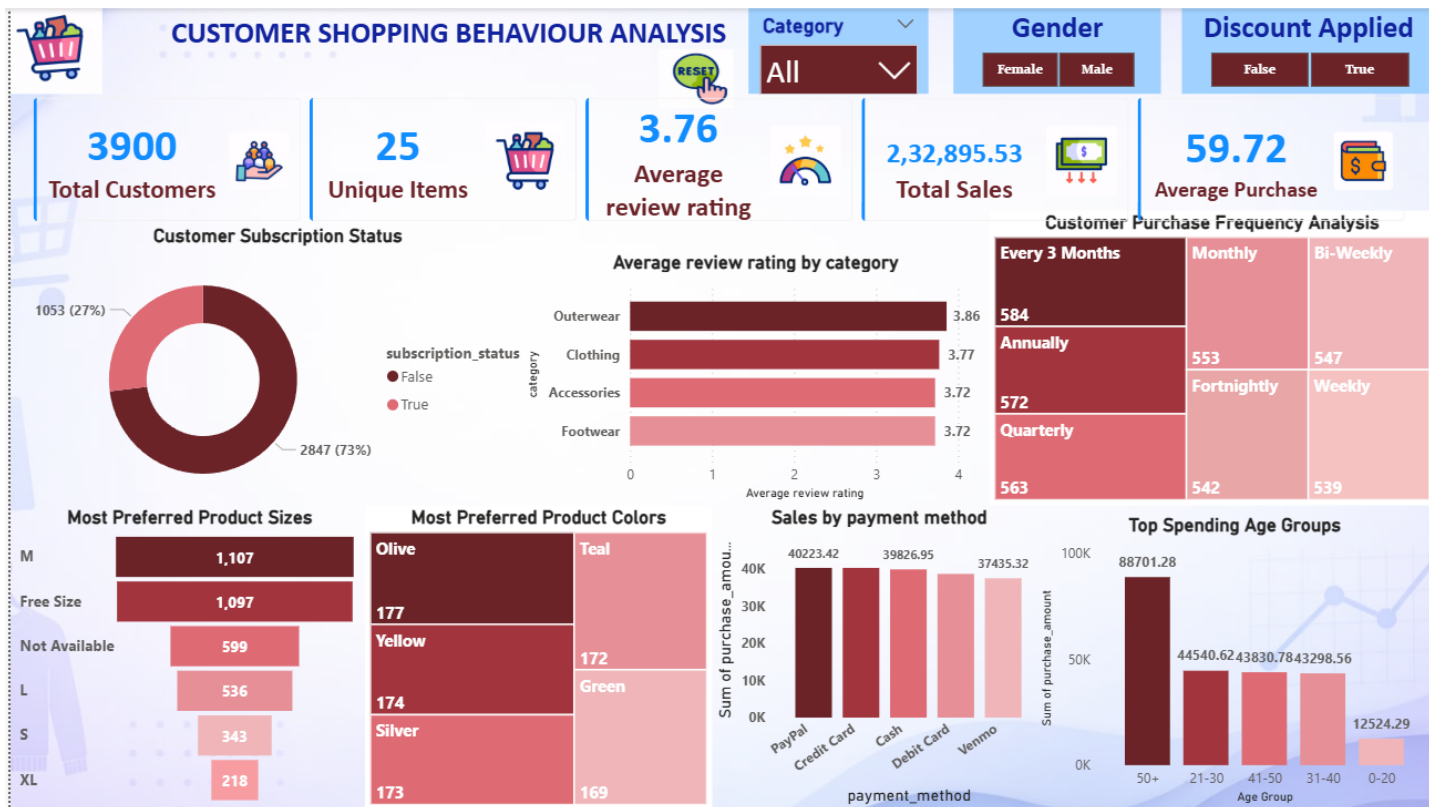
	total_sales numeric 🔒
1	232895.53

18. What is the average purchase amount per transaction?

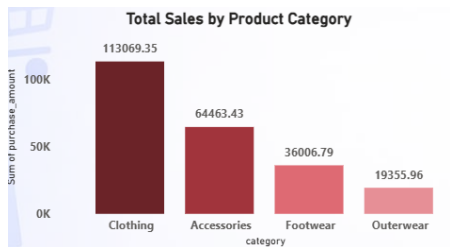
	avg_purchase	
	numeric	
1	59.72	

5. Power BI Dashboard

An interactive Power BI dashboard was created to visualize customer shopping behaviour and business performance. The dashboard contains charts and KPIs that help identify trends and business opportunities.



1. Total Sales by Product Category



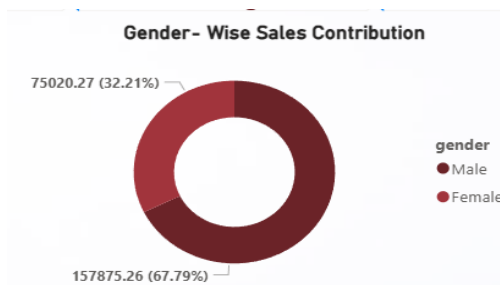
Business Insight

Clothing is the highest-selling category, showing strong customer demand and major contribution to total sales. Outerwear has the lowest sales performance.

Business Impact

The company should focus more on Clothing products to increase revenue and improve marketing strategies for low-performing categories like Outerwear.

2. Gender – Wise Sales Contribution



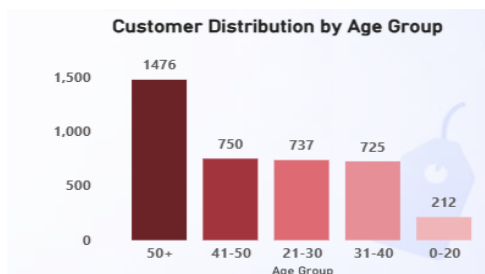
Business Insight

Male customers contribute the highest sales revenue, accounting for around 68% of total sales compared to female customers.

Business Impact

The business can target male customers with personalized marketing and product strategies while improving campaigns to increase female customer engagement and sales.

3. Customer Distribution by Age Group



Business Insight

Customers aged 50+ represent the largest customer group, while the 0–20 age group has the lowest customer count.

Business Impact

The business should focus more on older customers through targeted products and marketing strategies while creating attractive offers to engage younger customers.

4. Season – Wise Sales Trend



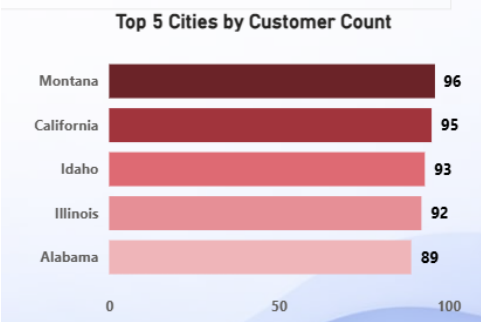
Business Insight

Sales are highest during the Fall season, while Summer records the lowest sales performance.

Business Impact

The business can maximize profits during high-sales seasons and introduce seasonal promotions or discounts in Summer to improve sales.

5. Top 5 Cities by Customer Count



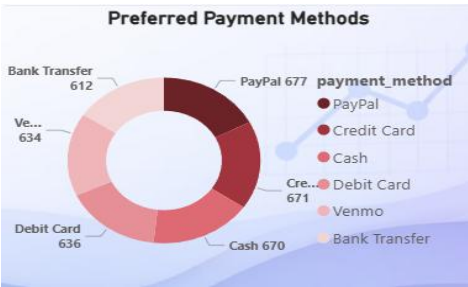
Business Insight

Montana and California have the highest customer count, showing strong customer presence in these locations.

Business Impact

The business can focus marketing campaigns and store expansion in high-performing locations to increase customer engagement and sales growth.

6. Preferred Payments Methods



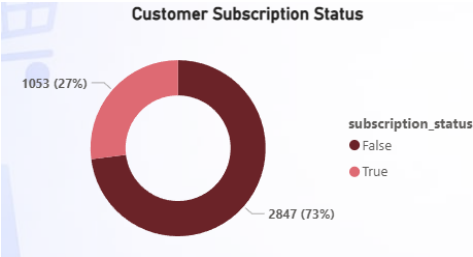
Business Insight

PayPal and Credit Card are the most preferred payment methods among customers.

Business Impact

The business should optimize digital payment options and offer payment-related rewards to improve customer convenience and increase sales.

7. Customer Subscription Status



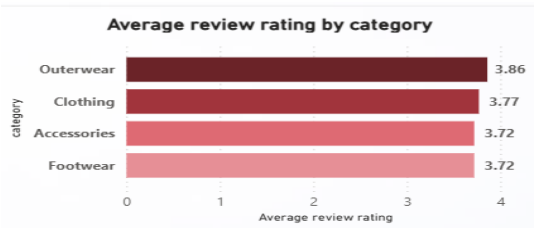
Business Insight

Most customers are non-subscribers, while only 27% customers have active subscriptions.

Business Impact

The business can increase customer loyalty and recurring sales by promoting subscription benefits and membership offers.

8. Average review rating by category



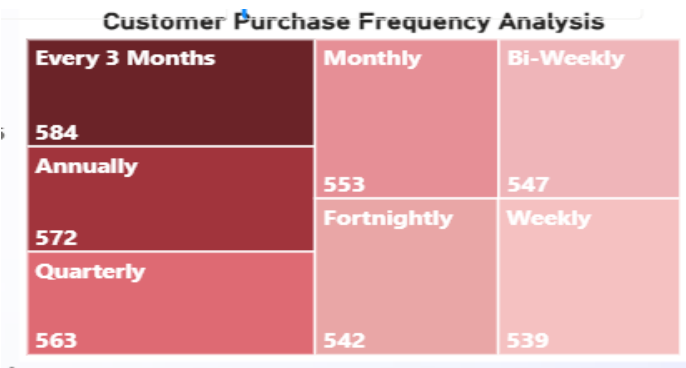
Business Insight

Outerwear has the highest average customer review rating, showing better customer satisfaction compared to other categories.

Business Impact

The business can maintain product quality in high-rated categories and improve customer experience in lower-rated categories to increase satisfaction and sales.

9. Customer Purchase Frequency Analysis



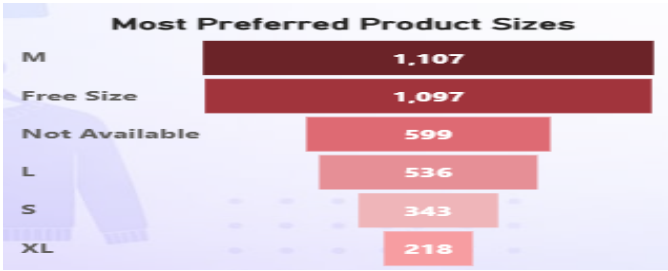
Business Insight

Most customers make purchases every 3 months, showing a moderate repeat purchase pattern.

Business Impact

The business can increase repeat purchases by offering loyalty programs, reminders, and personalized promotions to encourage more frequent shopping.

10. Most Preferred Product Sizes



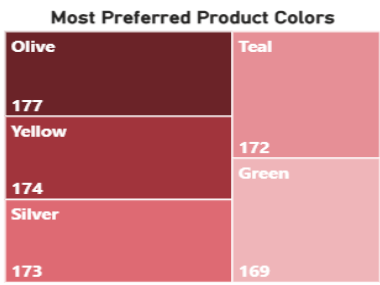
Business Insight

Medium (M) and Free Size products are the most preferred sizes among customers.

Business Impact

The business should maintain higher inventory for popular sizes to meet customer demand and reduce stock shortages.

11. Most Preferred Product Colors



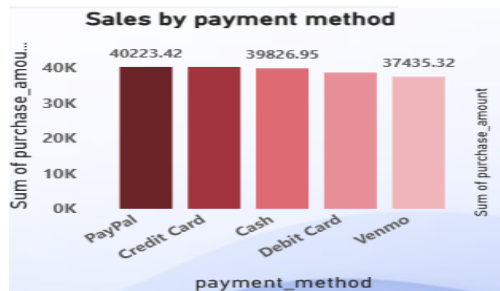
Business Insight

Olive and Yellow are the most preferred product colors among customers.

Business Impact

The business should stock more products in popular colors to match customer preferences and improve sales performance.

12. Sales by payment Method



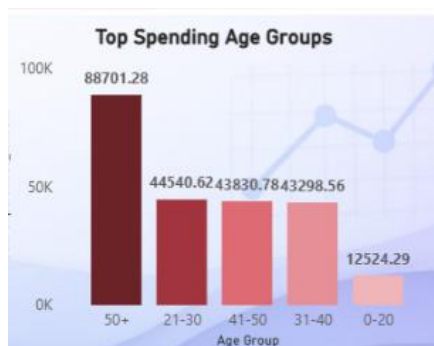
Business Insight

PayPal and Credit Card generate the highest sales revenue among all payment methods.

Business Impact

The business should strengthen digital payment options and offer payment-based rewards to improve customer experience and boost sales.

13. Top Spending Age Groups



Business Insight

Customers aged 50+ contribute the highest sales revenue, while the 0–20 age group contributes the least.

Business Impact

The business should focus marketing and premium product strategies on high-spending older customers to maximize revenue growth.

6. Key Business Insights

1. Clothing is the highest-selling category, showing strong customer demand and major contribution to total sales. Outwear has the lowest sales performance.
2. Male customers contribute the highest sales revenue, accounting for around 68% of total sales compared to female customers.
3. Customers aged 50+ represent the largest customer group, while the 0–20 age group has the lowest customer count.
4. Sales are highest during the Fall season, while Summer records the lowest sales performance.
5. Montana and California have the highest customer count, showing strong customer presence in these locations.
6. PayPal and Credit Card are the most preferred payment methods among customers.

7. Most customers are non-subscribers, while only 27% customers have active subscriptions.
8. Outerwear has the highest average customer review rating, showing better customer satisfaction compared to other categories.
9. Most customers make purchases every 3 months, showing a moderate repeat purchase pattern.
10. Medium (M) and Free Size products are the most preferred sizes among customers.
11. Olive and Yellow are the most preferred product colors among customers.
12. PayPal and Credit Card generate the highest sales revenue among all payment methods.
13. Customers aged 50+ contribute the highest sales revenue, while the 0–20 age group contributes the least.

7. Business Recommendations

1. Increase the stock of Clothing products because they have the highest sales and customer demand.
2. Improve Outerwear sales by offering discounts and attractive product designs.
3. Focus more on male customers because they generate the highest sales revenue.
4. Provide special offers and personalized products for customers aged 50+.
5. Use social media marketing to attract younger customers (0–20 age group).
6. Run Summer season offers to increase sales during low-performing months.
7. Expand marketing activities in Montana and California because these locations have the highest customer count.
8. Encourage more customers to take subscriptions by giving exclusive discounts and rewards.
9. Continue supporting PayPal and Credit Card payment methods because customers prefer them the most.
10. Keep more stock of Medium (M) and Free Size products because they are highly preferred by customers.
11. Increase the availability of Olive and Yellow colored products due to high customer preference.
12. Improve product quality in other categories by using the positive feedback from the Outerwear category.
13. Create loyalty and reward programs to encourage repeat purchases from customers.

8. Conclusion

This project successfully demonstrated the complete Data Analyst workflow including data cleaning, EDA, SQL analysis, and dashboard visualization. The project helped generate valuable business insights related to customer behaviour, sales trends, and purchasing patterns. The analysis can support better business decision-making, customer targeting, inventory planning, and marketing strategies.

9. Tools & Technologies Used

- Python (Pandas, NumPy)
- Jupyter Notebook
- SQLPower BI
- CSV Dataset